

12. How does the pitot tube help in measuring the air speed?
 - A) Using temperature
 - B) Using velocity
 - C) Using mach number
 - D) Using the pressure difference
13. As the ultrasonic frequency increases
 - A) Velocity of sound increases
 - B) Velocity of sound decreases
 - C) Wavelength decreases
 - D) Wavelength increases
14. What does Lidar stand for?
 - A) Light Detection and Ranging
 - B) Laser Imaging and Detection
 - C) Light Imaging and Ranging
 - D) Laser Detection and Ranging
15. What happens to the magnetic field in the solenoid when the number of turns increases?
 - A) Increases
 - B) Decreases
 - C) Remains constant
 - D) Becomes zero
16. Which motor type is commonly used in electric vehicles for regenerative braking?
 - A) DC series motor
 - B) Synchronous motor
 - C) Induction motor
 - D) Brushless DC motor
17. PWM stand for
 - A) pulse-width modulation
 - B) pulse with modem
 - C) pulse wide moderation
 - D) None of the mentioned
18. The direction of rotation of the D.C series motor can be changed by
 - A) Interchanging field terminal
 - B) Interchanging supply terminal
 - C) Interchanging field terminal & supply terminal
 - D) None of the mentioned
19. What type of fuse is commonly used in automotive applications due to its compact size and vibration resistance?
 - A) Blade fuse
 - B) Microfuse
 - C) High-rupture capacity fuse
 - D) Time-lag fuse
20. Inductor and Capacitor in the Buck converter are used to _____
 - A) Increase the cost
 - B) Decrease the cost
 - C) Filter out the harmonics
 - D) Increase the harmonics

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat diagram, explain the measuring principle of a strain-gauge-based pressure sensor.	8	L*2	1	1
	b) With a neat sketch, explain the measured variables in the vehicle coordinates system used in the Radar principle.	8	L2	2	1
2.	a) With a neat diagram, explain the working of differential magneto resistive sensors.	8	L2	1	1
	b) With a neat schematic representation, explain the video camera used in driver-assistance systems.	8	L2	2	1
3.	a) With a neat diagram, explain the measuring principle of the passive rpm sensor using the Hall effect used in magnetostatic sensors.	8	L2	1	1
	b) With a neat diagram, explain the basic principle of photo-sensing of photodiodes used in an image sensor.	8	L2	2	1
Unit – II					
4.	a) What are the differences between an intelligent switch and a solid-state relay?	8	L2	3	1
	b) With a neat diagram and waveforms, explain the working of single-phase full-bridge inverter.	8	L2	4	1
5.	a) Explain the working of variable reluctance motor with a neat diagram.	8	L2	3	1

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| b) | With a neat diagram, explain the unidirectional DC motor drive with power MOSFETs. | 8 | L2 | 4 | 1 |
| 6. a) | With a neat circuit diagram, explain the operation of open-loop control principle for a brushless dc motor. | 8 | L2 | 3 | 1 |
| b) | With relevant sketch, explain dead-time in the PWM generator circuits. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | With a neat circuit diagram and waveforms, explain the operation of a flyback converter operated in continuous conduction mode. | 8 | L2 | 5 | 1 |
| b) | With a neat diagram, explain the reverse voltage protection used in distribution bus using (i) Schottky diode (ii) P-type FET with a Zener clamp and (iii) smart diode. | 8 | L2 | 5 | 1 |
| 8. a) | What are the requirements and best practices for the electrical energy system design. | 8 | L2 | 5 | 1 |
| b) | With a neat circuit diagram and waveforms, explain the operation of direct (forward) converter. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
